CRISTIAN ALVARADO

Software Engineer | AI Systems & Distributed Software

Pereira, Colombia | English B2 (upper-intermediate)
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PROFILE

Software Engineer at Mercado Libre, building software and automation for Mercado Pago across Latin America. Experience combines AI agents, MCP servers, reusable skills and developer tooling with backend services and React/TypeScript interfaces. Machine Learning and Deep Learning instructor at Universidad Tecnologica de Pereira (UTP).

CURRENT EXPERIENCE

Software Engineer

Mercado Libre / Mercado Pago | Dec 2023 - Present

- Build AI agents, MCP integrations, reusable skills and automated workflows connected to engineering tools and services.

- Develop backend services and integrations with Go, Node.js and Java, and product interfaces with React and TypeScript.

- Work with databases, CI/CD, observability and regional solutions across Brazil, Mexico, Argentina, Colombia, Chile and Peru.

Machine Learning & Deep Learning Instructor

Universidad Tecnologica de Pereira | Sep 2026 - Present

Teach Machine Learning and Deep Learning in the Data Science diploma program, connecting model training and evaluation with applied problems.

- Created Nexo, a teaching tool connecting interactive explanations, practical Python/Colab exercises and learning progress.

FEATURED PROJECT

Nexo - a teaching tool I created

Practical notebooks and visual tools connecting preprocessing, training and evaluation with regression, classification, clustering, dimensionality reduction and computer vision.

Architecture: JavaScript, Three.js, Python/Colab and an authenticated records API backed by D1/SQLite. Guided demonstrations available by contact.

CRISTIAN ALVARADO

Software Engineer | Experience & technical practice

EARLIER EXPERIENCE

Machine Learning & Deep Learning Instructor

Universidad Tecnologica de Pereira | Sep - Oct 2025

Taught Machine Learning and Deep Learning in the Data Science diploma program, connecting learning algorithms and artificial neural networks with practical applications for professional learners.

Software Engineering Instructor & Speaker

CIAF Educacion Superior | Jan - Dec 2023

Taught programming and software engineering across multiple levels. Covered programming logic, architecture, agile methods and project management through practical learning activities.

Software Engineer

Universidad Tecnologica de Pereira | Sep 2020 - Oct 2023

- Developed and supported institutional software and academic systems.
- Redesigned interfaces to improve interaction and usability in academic applications.
- Led development of a modular CRM for institutional relationship management.

TECHNICAL PRACTICE

Applied AI: AI agents, Model Context Protocol (MCP), reusable skills, workflow automation and developer tooling.

Backend and product: Go, Node.js, Java, Python, TypeScript, React, APIs and databases.

Delivery and operations: CI/CD, AWS/GCP, Kubernetes, Terraform, observability and distributed software.

Teaching: Machine Learning, Deep Learning, model evaluation, practical notebooks and interactive technical explanations.

EDUCATION

Software Engineering - CIAF Educacion Superior.

PUBLIC WORK & CONTACT

Portfolio: kthyon.github.io/portafolio/
LinkedIn: linkedin.com/in/crisalvarado-ingeniero/